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United States Patent	12409003
Kind Code	B2
Date of Patent	September 09, 2025
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Instrument cassette assemblies for robotic surgical instruments

Abstract

A surgical instrument of a robotic surgical system includes an elongated shaft assembly, and end effector, and an instrument cassette assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The end effector is supported the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing and an actuator system supported in the cassette housing. The actuator system is operably coupled to the end effector for operating the end effector.

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Appl. No.: 17/686749

Filed: March 04, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220361971 A1	Nov. 17, 2022

Related U.S. Application Data

us-provisional-application US 63188554 20210514

Publication Classification

Int. Cl.: **A61B34/37** (20160101); **A61B34/00** (20160101); **A61B34/30** (20160101); **A61B34/35** (20160101); **A61B90/50** (20160101)

U.S. Cl.:

CPC **A61B34/37** (20160201); **A61B34/35** (20160201); **A61B34/71** (20160201); **A61B90/50** (20160201); A61B2034/301 (20160201); A61B2034/302 (20160201)

Field of Classification Search

USPC: None

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/188,554, filed May 14, 2021, the entire contents of which are incorporated by reference herein.

TECHNICAL FIELD

(1) This disclosure relates to robotic systems and, more particularly, to instrument cassettes for robotic surgical instruments.

BACKGROUND

(2) Surgical instruments used in laparoscopic and/or robotic surgery generally have a proximally located actuating mechanism that may be used to actuate a distal end effector for performing a surgical task within a body cavity of a patient. Such instruments may be used in applications where there is an area of limited access for an operator. The distal end of the instrument may be inserted into the area of limited access and the operator may remotely and/or robotically manipulate the instrument via the actuator mechanism.

SUMMARY

(3) In accordance with an aspect of this disclosure, a robotic surgical system includes a drive unit and a surgical instrument removably connected to the drive unit. The surgical instrument includes an elongated shaft assembly, an end effector, and an instrument cassette assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The end effector is supported at the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing, an actuator system supported in the cassette housing and operably coupled to the end effector for operating the end effector. The actuator system includes a cable actuator assembly, a shaft assembly defining a longitudinal axis, a rotation actuator assembly, and an axial actuator assembly. The cable actuator assembly includes a plurality of cables that extends from the cassette housing to the end effector for manipulating the end effector. The rotation actuator assembly is coupled to the shaft assembly and positioned to rotate the shaft assembly about the longitudinal axis for imparting rotational force to the end effector. The axial actuator assembly is coupled to the shaft assembly and positioned to axially translate the shaft assembly relative to the longitudinal axis for imparting axial force to the end effector.

(4) In aspects, the cable actuator assembly may include a crank, a first slider, and a second slider, the first and second sliders coupled to the crank. The crank may be rotatable to linearly translate the first and second sliders relative to one another. The first slider may support a first cable of the plurality of cables and the second slider may support a second cable of the plurality of cables. The crank may be coupled to a driver that is engaged with the drive unit. The driver may be configured to impart rotational force on the crank.

(5) In aspects, the rotation actuator assembly may include a drive wheel and a belt drive shaft supporting a belt. The belt may be coupled to the shaft assembly and the drive wheel may be coupled to the belt drive shaft. The drive wheel and the belt drive may be disposed transverse to one another. The drive wheel may be configured to rotate the belt drive shaft. Rotation of the belt drive shaft may rotate the belt to rotate the shaft assembly.

(6) In aspects, the axial actuator assembly may include a drive disc, a drive arm coupled to the drive disc, and a drive plate coupled to the drive arm and to the shaft assembly. In aspects, the drive arm may include a first pin coupled to the drive disc and a second pin coupled to the drive plate. The drive plate may define a pin slot that receives the second pin. The second pin may be slidable along the pin slot to axially translate the drive plate and the shaft assembly as the drive disc rotates.

(7) According to one aspect, this disclosure is directed to a surgical system including a cassette housing and an actuator system. The actuator system is supported in the cassette housing and includes a cable actuator, a shaft assembly, a rotation actuator assembly, and an axial actuator assembly. The cable actuator assembly includes a plurality of cables. The shaft assembly defines a longitudinal axis. The rotation actuator assembly is coupled to the shaft assembly and positioned to rotate at least a portion of the shaft assembly about the longitudinal axis. The axial actuator assembly is coupled to the shaft assembly and positioned to axially translate at least a portion of the

shaft assembly relative to the longitudinal axis.

(8) According to another aspect, this disclosure is directed to a surgical instrument for a robotic surgical system. The surgical instrument includes an elongated shaft assembly, an end effector, and an instrument cassette assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The end effector is supported at the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing and an actuator system. The actuator system is supported in the cassette housing and operably coupled to the end effector for operating the end effector. The actuator system includes a cable actuator assembly, a shaft assembly, a rotation actuator assembly, and an axial actuator assembly. The cable actuator assembly includes a plurality of cables that extends from the cassette housing to the end effector for manipulating the end effector. The shaft assembly defines a longitudinal axis. The rotation actuator assembly is coupled to the shaft assembly and positioned to rotate the shaft assembly about the longitudinal axis for imparting rotational force to the end effector. The axial actuator assembly is coupled to the shaft assembly and positioned to axially translate the shaft assembly relative to the longitudinal axis for imparting axial force to the end effector.

(9) According to still another aspect, this disclosure is directed to a robotic surgical system. The robotic surgical system includes a drive unit and a surgical instrument removably connected to the drive unit. The surgical instrument includes an elongated shaft assembly, an end effector, and an instrument cassette assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The end effector is supported on the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing and an actuator system. The actuator system is supported in the cassette housing and is operably coupled to the end effector for operating the end effector. The actuator system includes a cable actuator assembly including a spindle, an upper crank, and a lower crank. The upper crank is coupled to a first cable and the lower crank is coupled to a second cable. The upper and lower cranks are movable along the spindle to move the first and second cables for manipulating the end effector.

(10) In aspects, the first and second cables may be movable relative to one another.

(11) In aspects, the drive unit may rotate the spindle about a spindle axis. Rotation of the spindle may cause the upper and lower cranks to translate along the spindle axis. The upper and lower cranks may translate in opposite directions along the spindle axis.

(12) In aspects, the second cable may extend through the lower crank. The lower crank may include a spine through which the second cable slides as the upper crank moves relative to the lower crank.

(13) In aspects, the upper crank may define a first spiral passage through an outer surface thereof. The lower crank may define a second spiral passage through an outer surface thereof. The second spiral passage may turn in an opposite direction than the first spiral passage. The spindle may include a first pin that slides through the first spiral passage and a second pin that slides through the second spiral passage.

(14) According to another aspect, this disclosure is directed to a surgical system. The surgical system includes a cassette housing and an actuator system. The actuator system is supported in the cassette housing. The actuator system includes a cable actuator assembly including a spindle, an upper crank, and a lower crank. The upper crank is coupled to a first cable. The lower crank is coupled to a second cable. The upper and lower cranks are movable along the spindle to move the first and second cables.

(15) In aspects, the spindle may rotate about a spindle axis to cause the upper and lower cranks to translate along the spindle axis.

(16) According to still another aspect, this disclosure is directed to a surgical instrument for a robotic surgical system. The surgical instrument includes an elongated shaft assembly, an end effector, an instrument cassette assembly. The elongated shaft assembly has a proximal end portion

and a distal end portion. The end effector is supported the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing and a cable actuator assembly. The cable actuator assembly is supported in the cassette housing and includes a spindle, an upper crank, and a lower crank. The upper crank is coupled to a first cable. The lower crank is coupled to a second cable. The upper and lower cranks are translatable along the spindle to move the first and second cables for manipulating the end effector as the spindle rotates relative to the upper and lower cranks.

(17) According to yet another aspect, this disclosure is directed to a robotic surgical system. The robotic surgical system includes a drive unit and a surgical instrument removably connected to the drive unit. The surgical instrument includes an elongated shaft assembly, an end effector, and an instrument cassette assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The end effector is supported on the distal end portion of the elongated shaft assembly. The instrument cassette assembly is supported on the proximal end portion of the elongated shaft assembly. The instrument cassette assembly includes a cassette housing and an actuator system. The actuator system is supported in the cassette housing and operably coupled to the end effector for operating the end effector. The actuator system includes a cable actuator assembly and a drive actuator assembly. The cable actuator assembly includes a crank that supports an upper slider and a lower slider. The upper and lower sliders are coupled to cables that extend to the end effector. The drive actuator assembly includes a rotation actuator assembly and an axial actuator assembly. The rotation actuator assembly has at least one spool that rotates an inner shaft assembly coupled to the end effector to impart rotational force to the end effector. The axial actuator assembly includes a pivotable clevis that moves an axial drive cable relative to the inner shaft assembly to impart axial force to the end effector.

(18) In aspects, the crank may be coupled to the upper and lower sliders by first and second pins. The first pin may be slidably positioned within an elongated pin slot defined in the upper slider and the second pin may be slidably positioned within an elongated pin slot defined in the lower slider. The upper slider and lower slider may be positioned to translate in opposite directions as the crank rotates.

(19) In aspect, the at least one spool of the rotation actuator assembly may include an input spool and an output spool that are coupled together by a rotation cable. The input spool may be nonrotatably coupled to a driver, the input spool configured to rotate when the driver rotates. Rotation of the input spool moves the rotation cable about the output spool to rotate the inner shaft assembly.

(20) In aspects, the axial actuator assembly may include a threaded nut that is pinned to the pivotable clevis to enable the pivotable clevis to pivot relative to the threaded nut. The threaded nut may be threadedly coupled to a threaded driver. The threaded driver may be rotatable to cause the threaded nut to translate along the threaded driver. Translation of the threaded nut along the threaded driver may cause the pivotable clevis to pivot about a mounting protrusion such that the axial drive cable moves between extended and retracted positions relative to the inner shaft assembly.

(21) According to one aspect, this disclosure is directed to a surgical system. The surgical system includes a cassette housing and an actuator system supported in the cassette housing. The actuator system includes a cable actuator assembly and a drive actuator assembly. The cable actuator assembly includes a crank that supports an upper slider and a lower slider. The upper and lower sliders are coupled to cables. The drive actuator assembly includes a rotation actuator assembly and an axial actuator assembly. The rotation actuator assembly has at least one spool that rotates an inner shaft assembly. The axial actuator assembly includes a pivotable clevis that moves an axial drive cable relative to the inner shaft assembly.

(22) According to yet another aspect, this disclosure is directed to a surgical instrument for a

robotic surgical system. The surgical instrument includes an elongated shaft assembly, an end effector, a cassette housing, and a drive actuator assembly. The elongated shaft assembly has a proximal end portion and a distal end portion. The elongated shaft assembly includes an inner shaft assembly. The end effector is supported the distal end portion of the elongated shaft assembly. The cassette housing is supported on the proximal end portion of the elongated shaft assembly. The drive actuator assembly is supported in the cassette housing and is operably coupled to the end effector for operating the end effector. The drive actuator assembly includes a rotation actuator assembly and an axial actuator assembly. The rotation actuator assembly has a spool that rotates the inner shaft assembly to impart rotational force to the end effector. The axial actuator assembly includes a pivotable clevis that moves an axial drive cable relative to the inner shaft assembly to impart axial force to the end effector.

(23) Other aspects, features, and advantages will be apparent from the description, the drawings, and the claims that follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate aspects of this disclosure and, together with a general description of this disclosure given above, and the detailed description given below, explain the principles of this disclosure, wherein:
- (2) FIG. 1 is a perspective view of a robotic surgical system being used for a surgical procedure on a patient in accordance with the principles of this disclosure;
- (3) FIGS. 2-4 are progressive views illustrating surgical instruments of the robotic surgical system of FIG. 1 being manipulated within a body cavity of the patient;
- (4) FIG. 5 is an enlarged, perspective view of proximal portions of surgical instruments of one surgical instrument system of the robotic surgical system of FIG. 1;
- (5) FIG. 6 is an enlarged top view of FIG. 5;
- (6) FIG. 7 is a perspective view of one surgical instrument of the surgical instruments shown in FIG. 5;
- (7) FIG. 8 is an enlarged, perspective view of an instrument cassette assembly of the surgical instrument of FIG. 7 with portions thereof shown in phantom for clarity;
- (8) FIG. 9 is a perspective view, with parts separated, of the instrument cassette assembly of FIG. 8;
- (9) FIG. 10 is a perspective view, with parts separated, of an actuator system of the instrument cassette assembly of FIG. 9;
- (10) FIG. 11 is a perspective view of a crank of a cable actuator assembly of the actuator system of FIG. 10;
- (11) FIGS. 12 and 13 are progressive views illustrating cable actuator assemblies of the actuator system of FIG. 10 being actuated;
- (12) FIG. 14 is an enlarged view of the indicated area of detail shown in FIG. 8 and illustrating a rotation actuator assembly of the actuator system of FIG. 10 being actuated;
- (13) FIGS. 15-17 are progressive views illustrating an axial actuator assembly of the actuator system of FIG. 10 being actuated;
- (14) FIG. 18 is an enlarged, perspective view of proximal portions of surgical instruments of another surgical instrument system of the robotic surgical system of FIG. 1;
- (15) FIG. 19 is an enlarged top view of FIG. 18;
- (16) FIG. 20 is a perspective view, with parts separated, of another instrument cassette assembly of one of the surgical instruments of the surgical instrument system of FIG. 18, the instrument cassette

assembly including an outer housing assembly and an inner housing assembly;

(17) FIG. 21 is a perspective view, with parts separated, of the inner housing assembly of FIG. 20, the inner housing assembly including an actuator housing and an actuator assembly;

(18) FIG. 22 is a perspective view of the actuator assembly of FIG. 21;

(19) FIG. 23 is a perspective view, with parts separated, of the actuator assembly of FIG. 21, the actuator assembly shown with portions removed for clarity;

(20) FIG. 24 is an enlarged, cross-sectional view as taken along section line 24-24 shown in FIG. 20;

(21) FIG. 25 is an enlarged, cross-sectional view as taken along section line 25-25 shown in FIG. 21;

(22) FIGS. 26 and 27 are progressive views illustrating the actuator assembly of FIG. 21 being actuated;

(23) FIG. 28 is an enlarged, perspective view of proximal portions of surgical instruments of yet another surgical instrument system of the robotic surgical system of FIG. 1;

(24) FIG. 29 is an enlarged, top view of FIG. 28;

(25) FIG. 30 is a perspective view of one of the surgical instruments of FIG. 28;

(26) FIG. 31 is an enlarged, perspective view of an instrument cassette assembly of the surgical instrument of FIG. 30;

(27) FIG. 32 is an enlarged, perspective view of a cable actuator assembly of the instrument cassette assembly of FIG. 31;

(28) FIG. 33 is an enlarged, perspective view, with parts separated, of the cable actuator assembly of FIG. 32;

(29) FIGS. 34-36 are progressive views of the cable actuator assembly of FIG. 32 being actuated;

(30) FIG. 37 is a perspective view of the instrument cassette assembly of FIG. 31 with portions thereof shown in phantom for clarity;

(31) FIG. 38 is an enlarged perspective view, with parts separated, of a drive actuator assembly of the instrument cassette assembly of FIG. 31, the drive actuator assembly including an axial actuator assembly and a rotation actuator assembly, the rotation actuator assembly shown being actuated;

(32) FIG. 39 is a perspective view, with parts separated, of the drive actuator assembly of FIG. 38; and

(33) FIGS. 40-42 are progressive, cross-sectional views as taken along section line 40-40 shown in FIG. 37 and illustrating an actuation of the axial actuator assembly of the drive actuator assembly of FIG. 38.

DETAILED DESCRIPTION

(34) Aspects of this disclosure are described in detail with reference to the drawings, in which like reference numerals designate identical or corresponding elements in each of the several views. As used herein, the term “distal” refers to that portion of structure farther from the user, while the term “proximal” refers to that portion of structure, closer to the user. As used herein, the term “clinician” refers to a doctor, nurse, or other care provider and may include support personnel and/or equipment operators.

(35) In the following description, well-known functions or constructions are not described in detail to avoid obscuring the present disclosure in unnecessary detail.

(36) Robotic surgical systems have been used in minimally invasive medical procedures and can include robotic arm assemblies. Such procedures may be referred to as what is commonly referred to as “Telesurgery.” Some robotic arm assemblies include one or more robot arms to which surgical instruments can be coupled. Such surgical instruments include, for example, endoscopes, electrosurgical forceps, cutting instruments, staplers, graspers, electrocautery devices, or any other endoscopic or open surgical devices. Prior to or during use of the robotic surgical system, various surgical instruments can be selected and connected to the robot arms for selectively actuating end effectors of the connected surgical instruments.

(37) With reference to FIGS. 1-4, a robotic surgical system is shown generally at **10**. Robotic surgical system **10** employs various robotic elements to assist the clinician and allow remote operation (or partial remote operation) of surgical instruments **100, 200, 300** of surgical instrument systems **50, 60, 70** of robotic surgical system **10**. Various controllers, circuitry, robotic arms, gears, cams, pulleys, electric and mechanical motors, etc. may be employed for this purpose and may be designed with surgical system **10** to assist the clinician during an operation or treatment. Such robotic systems may include remotely steerable systems, automatically flexible surgical systems, remotely flexible surgical systems, remotely articulating surgical systems, wireless surgical systems, modular or selectively configurable remotely operated surgical systems, etc.

(38) Robotic surgical system **10** includes a workstation **12** and an instrument cart **14**. The instrument cart **14** includes one or more surgical instrument systems **50, 60, 70** mounted on a moveable drive unit **18** that houses an instrument drive assembly **20** for manipulating the surgical instrument systems **50, 60, 70** and/or independent surgical instruments **100, 200, 300** thereof with the assistance of, for example one or more computing devices or controllers. The surgical instruments **100, 200, 300** can include, for example, graspers or forceps **26**, which may be electrosurgical, an endoscope **28**, and/or any other suitable instrument that can be driven by one or more associated tool drives (not shown) of instrument drive assembly **20**. For example, besides graspers **26** and endoscope **28**, the one or more surgical instruments **100, 200, 300** can include dexterous tools, such as grippers, needle drivers, staplers, dissectors, cutters, hooks, graspers, scissors, coagulators, irrigators, suction devices, that are used for performing a surgical procedure.

(39) Each surgical instrument system **50, 60, 70** includes an insertion tube **16** defining a plurality of separate conduits, channels or lumens **16a** therethrough that are configured to receive, for instance, the surgical instruments **100, 200, 300** for accessing a body cavity “BC” of a patient “P.” In other aspects, the insertion tube **16** may define a single conduit, channel or lumen therethrough that is configured to receive, for instance, the surgical instruments **100, 200, 300** for accessing a body cavity “BC” of a patient “P.” In particular, the insertion tube **16** can be inserted through an incision “I” and/or access device **17** (e.g., a surgical portal, which may include one or more seals to facilitate sealed insertion through tissue “T” of the patient “P”) and into the body cavity “BC” of the patient “P”). With insertion tube **16** positioned in the patient “P,” the surgical instruments **100, 200, 300** can be advanced through insertion tube **16** into the body cavity “BC” of the patient “P.” Further, the workstation **12** includes an input device **22** for use by a clinician for controlling the insertion tube **16** and the various surgical instrument systems **50, 60, 70** (and surgical instruments **100, 200, 300** thereof) via the instrument drive assembly **20** to perform surgical operations on the patient “P” while the patient “P” is supported on a surgical table **24**, for example. Input device **22** is configured to receive input from the clinician and produces input signals. Input device **22** may also be configured to generate feedback to the clinician. The feedback can be visual, auditory, haptic, or the like.

(40) The workstation **12** can further include computing devices and/or controllers such as a master processor circuit **22a** in communication with the input device **22** for receiving the input signals and generating control signals for controlling the robotic surgical system **10**, which can be transmitted to the instrument cart **14** via an interface cable **22b**. In some cases, transmission can be wireless and interface cable **22b** may not be present. The input device **22** can include right and left-hand controls (not shown) and/or foot pedals (not shown), which are moved/operated to produce input signals at the input device **22** and/or to control robotic surgical system **10**. The instrument cart **14** can include a slave processor circuit **20a** that receives and the control signals from the master processor circuit **22a** and produces slave control signals operable to control the various surgical instrument systems **50, 60, 70** (and surgical instruments **100, 200, 300** thereof) during a surgical procedure. The workstation **12** can also include a user interface, such as a display (not shown) in communication with the master processor circuit **22a** for displaying information (such as, body cavity images) for a region or site of interest (for example, a surgical site, a body cavity, or the like)

and other information to a clinician. While both master and slave processor circuits are illustrated, in other aspects, a single processor circuit may be used to perform both master and slave functions. (41) Turning now to FIGS. 5-17, surgical instrument system **50** of robotic surgical system **10** includes insertion tube **16** and a plurality of surgical instruments **100** that is insertable through insertion tube **16**. Although only three surgical instruments **100** are shown, surgical instrument system **50** can include any number and/or type of surgical instruments such as graspers **26** and endoscope **28** as noted above.

(42) As seen in FIGS. 6 and 7, surgical instrument **100** of surgical instrument system **50** defines a longitudinal axis “L” and includes an instrument cassette assembly **102** on a proximal end portion thereof, an elongated shaft assembly **104** that extends distally from instrument cassette assembly **102**, and an end effector **106** supported on a distal end portion of elongated shaft assembly **104**. End effector **106** is actuatable by instrument cassette assembly **102** for effectuating a surgical procedure. Indeed, actuating end effector **106** can cause end effector **106** to, for example, articulate, pivot, clamp, rotate, etc. relative to the longitudinal axis “L” of surgical instrument **100** for repositioning end effector **106** and/or for treating tissue “T” of the patient “P” as noted above (see FIGS. 2-4).

(43) With reference to FIG. 8, instrument cassette assembly **102** of surgical instrument **100** includes a cassette housing **108** that supports an actuator system **110** and is coupled to elongated shaft assembly **104**. Actuator system **110** includes a plurality of cable actuator assemblies **112**, a rotation actuator assembly **114**, and an axial actuator assembly **116**.

(44) As seen in FIG. 9, cassette housing **108** of instrument cassette assembly **102** includes a first housing **108a**, a second housing **108b**, and a lid **108c** that couple together to support the actuator system **110** therein and secure cassette housing **108** to elongated shaft assembly **104**.

(45) With reference to FIGS. 10-13, each cable actuator assembly **112** of the actuator system **110** of instrument cassette assembly **102** includes a crank **112a** that supports a bearing **112b** and a driver **112c** on a first end of the crank **112a**. The crank **112a** further supports a first slider **112d** and a second slider **112e** on a second end of the crank **112a**. The first slider **112d** is supported on a first side of the crank **112a** and the second slider **112e** is supported on a second side of the crank **112** opposite to the first side of the crank **112a**. The first slider **112d** supports a first cable **112g** and the second slider **112e** supports a second cable **112f**. The first slider **112d** defines a first slot **112h** and the second slider **112e** defines a second slot **112k**. The crank **112a** includes a plate **112m** having a central bearing prong **112n** extending from the first end thereof. The central bearing prong **112n** is received through the bearing **112b** and is nonrotatably coupled to driver **112c** of the cable actuator assembly **112**. In aspects, central bearing prong **112n** of the crank **112a** can be keyed to a bore **112x** defined in driver **112c** on the second end of driver **112c** to enable crank **112a** and driver **112c** to be press-fit together. Central bearing prong **112n** of crank **112a** and bore **112x** of driver **112c** can have any suitable counterpart geometry (e.g., square, triangle, star, chamfer, bevel, fillet, edge, groove, etc.) to enable driver **112c** to impart rotational driving force to crank **112a** via the nonrotatable coupling of driver **112c** and crank **112a**. In aspects, driver **112c** can be secured to crank **112a** via any suitable technique such as sonic welding, adhesive, fastener, snap-fit, etc., or combinations thereof such that driver **112c** can rotate crank **112a** about a central pivot axis “P1.” Briefly, as seen in FIG. 5, driver **112c** defines a drive slot **112y** therein to receive rotational drive force from moveable drive unit **18**.

(46) With continued reference to reference to FIGS. 10-13, the plate **112m** of crank **112a** further includes a shoulder **112p** having a first finger **112q** extending from a second side of the plate **112m**, and a second finger **112r** extending from the first side of the plate **112m** and recessed from the first finger **112q**. The first and second fingers **112q**, **112r** of crank **112a** are received within first and second slots **112h**, **112k** of the respective first and second sliders **112d**, **112e**. Crank **112a** is configured to move (e.g., translation or linear movement, which may be reciprocating movement) first and second sliders **112d**, **112e** relative to one another, as indicated by arrows “F” and “G” as

shown in FIG. 13, when crank **112a** rotates in clockwise and/or counterclockwise directions as indicated by arrows “H” and “I.” Linear movement of first and second sliders **112d**, **112e** causes first and second cables **112f**, **112g** to actuate (e.g., articulate, elevate, fire, clamp, etc. end effector **106** and/or jaw members thereof). Each cable actuator assembly **112** of the actuator system **110** may move independent and/or dependent of one or more of the other cable actuator assemblies **112** to actuate/operate the end effector **106** as desired.

(47) Referring to FIGS. 10-14, rotation actuator assembly **114** of the actuator system **110** includes a driver **114a**, a bearing **114b**, a drive wheel **114c**, a belt **114d**, a belt drive shaft **114e**, and a belt drum **114f**. Drive wheel **114c** of rotation actuator assembly **114** includes a first bevel gear **114g**. Belt drive shaft **114e** of rotation actuator assembly **114** includes a second bevel gear **114h** that is transverse to first bevel gear **114g** and positioned to meshingly engage first bevel gear **114g** of drive wheel **114c**. Belt drive shaft **114e** further includes a drum gear **114y** supported adjacent to second bevel gear **114h** and positioned to enable belt **114d** to slide therealong as belt **114d** rotates about belt drive shaft **114e**. First bevel gear **114g** is configured to rotate second bevel gear **114h** about pin axis “P2,” as indicated by arrow “R1,” when first bevel gear **114g** rotates about drive axis “D1,” as indicated by arrow “R2.” Rotation of second bevel gear **114h** about drive axis “D1,” as indicated by arrow “R1,” causes belt **114d** to rotate about belt drive shaft **114e** and belt drum **114f**, as indicated by arrow “R3.”

(48) Belt drum **114f** of rotation actuator assembly **114** is connected to a shaft assembly **115** including an outer shaft **115a** and an inner shaft assembly **115b** such that rotation of belt **114d** causes belt drum **114f** to rotate about shaft axis “A1” defined by shaft assembly **115**, as indicated by arrow “R4.” Inner shaft assembly **115b** includes a first inner shaft **115c** and a second inner shaft **115d** that is slidably advanceable through first inner shaft **115c** along shaft axis “A1” of shaft assembly **115**. First inner shaft **115c** supports belt drum **114f** on a first end thereof with the second end of first inner shaft **115c** coupled to end effector **106** (FIG. 7) for imparting rotational movement/force on end effector **106**. Belt drum **114f** is positioned to rotate about shaft axis “A1” as belt **114d** rotates about belt drum **114f**. Further, like driver **112c** of cable actuator assembly **112**, driver **114a** of rotation actuator assembly **114** includes a drive slot **112y** on a first end thereof. Likewise, driver **114a** of rotation actuator assembly **114** is nonrotatably coupled to drive wheel **114c**, for example, via a central bearing prong **114n** extending from a first end of drive wheel **114c**. Central bearing prong **114n** supports bearing **114b** and may be mechanically coupled to the second end of driver **114a** via a bore **114x** of driver **114a** (e.g., press-fit) and/or via sonic welding, adhesive, fastener, etc., or combinations thereof such that driver **114a** imparts rotational movement on drive wheel **114c** about drive axis “D1” as driver **114a** rotates about drive axis “D1.”

(49) Referring to FIGS. 10-17, axial actuator assembly **116** of actuator system **110** includes a driver **116a**, a bearing **116b**, a drive disc **116c** defining an elongated pin notch **116d**, a drive arm **116e** coupled to drive disc **116c**, and a drive plate **116f** coupled to drive arm **116e**. Drive arm **116e** includes a first pin **116g** on a first end thereof that is slidably and rotatably received within elongated pin notch **116d** of drive disc **116c** to enable rotation of drive arm **116e** about pivot axis “P3” defined through first pin **116g**. Drive arm **116e** further includes a second pin **116j** on a second end thereof that is slidably and rotatably received within a pin slot **116k** defined through drive plate **116f** and disposed at an acute angle (e.g., 45 degrees) relative to shaft axis “A1.” Drive arm **116e** is also configured to pivot about pin axis “P4” defined through second pin **116j** as second pin **116j** slides through pin slot **116k** of drive plate **116f**. Further, like driver **112c** of cable actuator assembly **112**, driver **116a** of axial actuator assembly **116** includes a drive slot **112y** on a first end thereof. Likewise, driver **116a** of axial actuator assembly **116** is nonrotatably coupled to drive disc **116c**, for example, via a central bearing prong **116n** extending from a first end of drive disc **116c**. Central bearing prong **116n** supports bearing **116b** and may be mechanically coupled to the second end of driver **116a** via a bore **116x** of driver **116a** (e.g., press-fit) and/or via sonic welding, adhesive, fastener, etc., or combinations thereof such that driver **116a** imparts rotational movement on drive

disc **116c** about drive axis “D2” of axial actuator assembly **116** as driver **116a** rotates about drive axis “D2.”

(50) As seen in FIGS. **15-17**, axial actuator assembly **116** of actuator system **110** is positioned to move between an intermediate position (FIG. **15**), a retracted position (FIG. **16**), and an extended position (FIG. **17**). In the intermediate position, drive arm **116e** is parallel to belt **114d** and orthogonal to shaft axis “A1” with first pin **116g** supported on a first side of elongated pin notch **116d** and second pin **116j** substantially centered along pin slot **116k** of drive plate **116f**. Rotation of driver **116a** in a first direction (e.g., clockwise), as indicated by arrow “C,” causes first pin **116g** of drive arm **116e** to slide to a second side (e.g., proximally) of elongated pin notch **116d**, as indicated by arrow “S1.” Rotation of driver **116a** in the first direction also causes second pin **116j** of drive arm **116e** to slide to a first side of pin slot **116k**, such that drive plate **116f** is urged toward the extended position (e.g., distally), as indicated by arrow “S2,” to impart distal axial movement to shaft assembly **115** and distal axial force and/or movement to end effector **106**. Rotation of driver **116a** in a second direction (e.g., counterclockwise), as indicated by arrow “CC,” causes second pin **116j** of drive arm **116e** to slide to a second side (e.g., distally) of pin slot **116k** such that drive plate **116f** is urged toward the retracted position (e.g., proximally), as indicated by arrow “S3” to impart proximal axial movement to shaft assembly **115** and proximal axial force and/or movement to end effector **106**.

(51) Turning now to FIGS. **18-27**, surgical instrument system **60** of robotic surgical system **10** includes insertion tube **16** and a plurality of surgical instruments **200** that is insertable through insertion tube **16**. Although only three surgical instruments **200** are shown, surgical instrument system **60** can include any number and/or type of surgical instruments such as graspers **26** and endoscope **28** as noted above.

(52) As seen in FIG. **18**, surgical instrument **200** of surgical instrument system **60** defines a longitudinal axis “L2” and includes an instrument cassette assembly **202** on a proximal end portion thereof and an elongated shaft assembly **204** that extends from instrument cassette assembly **202** to an end effector **106** (FIG. **7**) supported on a distal end portion of elongated shaft assembly **204**. End effector **106** is actuatable by instrument cassette assembly **202** for effectuating a surgical procedure. Indeed, actuating end effector **106** can cause end effector **106** to, for example, articulate, pivot, clamp, rotate, etc. relative to the longitudinal axis “L2” of surgical instrument **200** for repositioning end effector **106** and/or for treating tissue “T” of the patient “P” as noted above (see FIGS. **2-4**).

(53) With reference to FIGS. **20** and **21**, instrument cassette assembly **202** of surgical instrument **200** includes an outer housing assembly **208** and an inner housing assembly **210** supported within outer housing assembly **208**. Inner housing assembly **210** includes an actuator housing **212** and an actuator assembly **214** that is supported within inner housing assembly **210**. Actuator housing **212** defines a plurality of actuator cavities **212a** defined therein for receiving actuator assembly **214**.

(54) Turning now to FIGS. **20-24**, actuator assembly **214** of inner housing assembly **210** includes a first set of cable actuator assemblies **216**, a second set of cable actuator assemblies **218**, and an axial actuator assembly **220** that are secured to a support plate **222**. Notably, adjacent cable actuator assemblies **216**, **218** may be disposed out of phase and/or offset from one another by, for example, 90 degrees (e.g., orthogonal to one another), and with axial actuator assembly **220** centered between first and second sets of cable actuator assemblies **216**, **218** to conserve space and reduce size requirements of inner housing assembly **210**. Support plate **222** defines bores **222a** for supporting cable and actuator assemblies **216**, **218**, and **220** therein, and cable passages **222b** for receiving cables **224** of cable actuator assemblies **216**, **218** therethrough. Although the first set of cable actuator assemblies **216** are shown to be longer than second set of cable actuator assemblies **218**, and the second set of cable actuator assemblies **218** is otherwise substantially the same as the first set of cable actuator assemblies **218**. Indeed, the first and second set of cable actuator assemblies **216**, **218** may have any suitable length. Axial actuator assembly **220** is coupled to a drive cable **226** that extends through a tube **228** extending from support plate **222** that couples to

elongated shaft assembly **204**.

(55) Each cable actuator assembly **216**, **218** of actuator assembly **214** includes a spindle **230**, an upper crank **232**, a lower crank **234**, an upper pin **236a**, a lower pin **236b**, an upper bearing **238a**, and a lower bearing **238b**.

(56) Spindles **230** of actuator assemblies **216**, **218** include an upper peg **230a** extending from a first end thereof and a lower peg **230b** extending from a second end thereof. Upper and lower pegs **230a**, **230b** secure to upper and lower bearings **238a**, **238b**, respectively. Upper peg **230a** is engageable with movable drive unit **18** to enable movable drive unit **18** to impart rotational drive force on spindles **230** (e.g., through drive couplers—not shown—of drive unit **18**). Each spindle **230** further defines an upper pin passage **230c** and a lower pin passage **230d** that extend transversely through spindle **230** at longitudinally spaced-apart locations and are positioned to receive upper and lower pins **236a**, **236b**, respectively, in a transverse (e.g., an orthogonal) relationship with spindle **230**.

(57) Lower crank **234** of each cable actuator assembly **216**, **218** defines a spindle passage **234a** longitudinally and centrally therethrough for receiving spindle **230** therethrough. Lower crank **234** further defines a spiral channel **234b** in an outer surface thereof. Spiral channel **234b** slidably receives lower pin **236b** of spindle **230** to enable lower crank **234** to axially slide along a lower portion of spindle **230**, as indicated by arrows “AA” (e.g., distally) and “AB” (e.g., proximally) when spindle **230** rotates about spindle axis “SA” relative to lower crank **234**, as indicated by arrows “RA” (e.g., clockwise) and “RB” (e.g., counterclockwise) shown in FIGS. **26** and **27**. Lower crank **234** further includes a spine **234c** that extends longitudinally along the outer surface of lower crank **234**. Spine **234c** defines cable channels **234d**, **234e** longitudinally therethrough that support cables **224**. A first cable **224a** of cables **224** is secured to lower crank **234** (e.g., via cable channel **234e**) and translates in the same direction and simultaneously with lower crank **234** (e.g., as indicated by arrows “AA” and “AB”). A second cable **224b** of cables **224** is slidably movable through cable channel **234d** of lower crank **234** and relative to lower yoke **234** as upper crank **232** translates relative to spindle **230**. Second cable **224b** is secured to upper crank **232** and movable with upper crank **232**.

(58) Upper crank **232** of each cable actuator assembly **216**, **218** is substantially like lower crank **234** but includes a spiral channel **232a** that turns along the outer surface thereof in an opposite direction as compared to spiral channel **234b** of lower crank **234**. And spiral channel **232a** of upper crank **232** slidably receives upper pin **236a** of spindle **230** to axially slide upper crank **232** along an upper portion of spindle **230**, as indicated by arrows “BA” (e.g., proximally) and “BB” (e.g., distally), when spindle **230** rotates about spindle axis “SA” and relative to upper crank **232**, as indicated by arrows “RA” (e.g., clockwise) and “RB” (e.g., counterclockwise) shown in FIGS. **26** and **27**. Upper and lower cranks **234**, **236** are positioned to translate in opposite axial directions relative to one another along spindle axis “SA” as upper and lower cranks **234**, **236** rotate about spindle **230** and spindle axis “SA.” Notably, upper crank **232** further includes a spine **232x** that supports and is secured to second cable **224b** to enable second cable **224b** to translate with upper crank **232** and relative to lower crank **234**, as indicated by arrows “BA” and “BB.” Upper crank **232** is separate and disconnected from first cable **224a**.

(59) Axial actuator assembly **220** is substantially like cable actuator assemblies **216**, **218**, but includes a spindle **230**, an upper crank **232**, an upper pin **236a**, an upper bearing **238a**, and a lower bearing **238b** (e.g., there is no lower crank or lower pin). Upper crank **232** is coupled to drive cable **226** and axially translatable upon rotation of spindle **230** thereof to translate drive cable **226** and impart axial drive force through drive cable **226** to, for example, end effector **106**.

(60) Turning now to FIGS. **28-42**, surgical instrument system **70** of robotic surgical system **10** includes insertion tube **16** and a plurality of surgical instruments **300** that is insertable through insertion tube **16**. Although only three surgical instruments **300** are shown, surgical instrument system **70** can include any number and/or type of surgical instruments such as graspers **26** and

endoscope **28** as noted above.

(61) As seen in FIG. **30**, surgical instrument **300** of surgical instrument system **70** defines a longitudinal axis “L3” and includes an instrument cassette assembly **302** on a proximal end portion thereof and an elongated shaft assembly **304** that extends from instrument cassette assembly **302** to an end effector **306** (FIG. **7**) supported on a distal end portion of elongated shaft assembly **304**. End effector **306** is actuatable by instrument cassette assembly **302** for effectuating a surgical procedure. Indeed, actuating end effector **306** can cause end effector **306** to, for example, articulate, pivot, clamp, rotate, etc. relative to the longitudinal axis “L3” of surgical instrument **300** for repositioning end effector **306** and/or for treating tissue “T” of the patient “P” as noted above with respect to end effector **106** (see FIGS. **2-4**).

(62) With reference to FIGS. **28-30**, instrument cassette assembly **302** of surgical instrument **300** includes an outer housing assembly **308** that supports an ID board **310**, a latch release mechanism **312** having a button release **312a** for selectively removing surgical instrument from movable drive unit **18**, and an electrosurgical socket **314** for selectively connecting to an electrosurgical energy source via an electrosurgical cable (not shown). Instrument cassette assembly **302** further includes an actuator system **316** supported within outer housing assembly **308** for actuating end effector **306**.

(63) Turning now to FIGS. **31-36**, actuator system **316** includes a plurality of cable actuator assemblies **318** and a drive actuator assembly **320**. The drive actuator assembly **320** includes an axial actuator assembly **322** and a rotation actuator assembly **324**.

(64) With reference to FIG. **32**, each cable actuator assembly **318** of the actuator system **316** includes a crank **330**, an upper slider **332**, a lower slider **334**, and a driver **336**. The upper slider **332** is secured to an upper surface of crank body **330x**, via a first pin **333a**, on a first side of crank body **330x**, the lower slider **334** is secured to a lower surface of crank body **330x**, via a second pin **333b**, on a second side of crank body **330x**. Crank body **330x** defines a first opening **330a** on the first side of crank body **330x** that receives a lower portion of the first pin **333a** and a second opening **330b** on the second side of crank body **330x** that receives an upper portion of the second pin **333b**. Upper slider **332** defines an elongated pin slot **332a** that receives an upper portion of the first pin **333a** and lower slider **334** defines an elongated pin slot **334a** that receives a lower portion of the second pin **333b**. Upper and lower sliders **332**, **334** further define ferrule openings **335** that support ferrules (not shown) therein for securing cables **338** of cable actuator assembly **318** to respective upper and lower sliders **332**, **334**. Crank **330** further includes a drive shaft **330c** that extends from the upper and lower surfaces of crank body **330x** and nonrotatably supports driver **336** so that rotation of driver **336** imparts rotational force to crank **330**.

(65) In some aspects, each cable actuator assembly **318** may be provided in the form of a rack and pinion arrangement. For example, crank **330** may be a pinion, and upper slider **332** and lower slider **334** are in the form of racks so that teeth of these respect rack and pinion feature engage one another. Indeed, upper and lower sliders **332**, **334** may be disposed in the same plane as one another (e.g., vertically aligned or in registration), and/or vertically offset from one another such that one is higher and/or lower than the other along a vertical or central axis (not explicitly shown) extending through a center of crank **330**.

(66) As seen in FIGS. **34-36**, rotation of driver **336** in a first direction (e.g., counterclockwise), as indicated by arrow “RCC,” rotates crank **330** such that first pin **333a** moves inwardly along elongated pin slot **332a** of upper slider **332** to an actuated position and second pin **333b** moves inwardly along elongated pin slot **334a** of lower slider **334** to an actuated position. As crank **330** is rotated in the first direction, crank **330** and first pin **333a** cause upper slider **332** to translate distally as indicated by arrow “DCC” while crank **330** and second pin **333b** cause lower slider **334** to translate proximally as indicated by arrow “PCC.” Rotation of driver **336** in a second direction opposite to the first direction (e.g., clockwise), as indicated by arrow “RC,” causes crank **330** to rotate such that first and second pins **333a**, **333b** move to the actuated position within respective

elongated pin slots **332a**, **334a** of upper and lower sliders **332**, **334** and drive upper slider **332** in a proximal direction, as indicated by arrow “PC,” and lower slider **334** in a distal direction, as indicated by arrow “DC.” As upper and lower sliders **332**, **334** translate between proximal and distal positions, cables **338** secured thereto translate with the respective upper and lower sliders **332**, **334**. For instance, when upper slider **332** translates distally (or proximally), a first cable **338a** of cables **338** secured to upper slider **332** translates distally (or proximally) with upper slider **332**, and vice versa. Similarly, when lower slider **334** translates proximally (or distally), a second cable **338b** of cables **338** secured to lower slider **334** translates proximally (or distally) with lower slider **334**, and vice versa.

(67) With reference to FIGS. **37-42**, drive actuator assembly **320** of actuator system **316** includes an actuator housing **321** having a first housing portion **321a** and a second housing portion **321b** that define actuator mounts **321c**, **321d** for supporting the axial actuator assembly **322** and the rotation actuator assembly **324** between the first and second housing portions **321a**, **321b**. Second housing portion **312b** further includes a mounting protrusion **312e** extending from a sidewall thereof.

(68) Rotation actuator assembly **324** of drive actuator assembly **320** includes an input spool **340**, an output spool **342**, a rotation cable **344** that couples to (e.g., wraps around) input and output spools **340**, **342**, bearings **346a**, **346b**, and a driver **348** that nonrotatably couples to input spool **340**. Cable **344** may have any number of windings about input and output spools **340**, **342** to enable rotational force to be transferred between input and output spools **340**, **342**. Notably, elongated shaft assembly **304** of surgical instrument **300** includes an inner shaft **304a** to which output spool **342** nonrotatably couples, and which is coupled to actuator housing **321** by a shaft bearing **343**. Output spool **342** imparts rotational force to inner shaft **304a** from input spool **340** as rotation cable **344** rotates output spool **342** about inner shaft axis “ISA,” as indicated by arrow “OS,” (as rotation cable **344** translates-see arrows “T1” and “T2”) in response to rotation of driver **348** about driver axis “DA,” as indicated by arrow “IS.”

(69) Axial actuator assembly **322** of drive actuator assembly **320** includes a clevis **350**, a threaded nut **352** mounted to clevis **350** via pins **352a** thereof, an upper bearing **354**, a lower bearing **356**, a threaded driver **357** having threads **357a**, and a cable pivot **358**. Threads **357a** of threaded driver **357** are threadedly engageable with threads **352b** of threaded nut **352** to enable threaded nut **352** to translate along threaded driver **357**, as indicated by arrows “N1” and “N2,” when threaded driver **357** is rotated, as indicated by arrows “TD1” and “TD2” (e.g., clockwise and/or counterclockwise about axis “TDA.” Clevis **350** defines a nut mount **350a** on a first end thereof that defines pin holes **350b** therethrough for receiving pins **352a** of threaded nut **352** therein. Clevis **350** further defines a protuberance hole **350c** that receives mounting protrusion **312e** of second housing portion **312b** for securing clevis **350** to actuator housing **321**. A second end of clevis **350** defines a cable pivot channel **350d** for receiving cable pivot **358** and an axial drive cable **360** therein. Axial drive cable **360** is coupled to cable pivot **358** on a first end thereof and extends through inner shaft **304a** to enable a second end of axial drive cable **360** to secure to end effector **306** for imparting axial drive force on end effector **306**.

(70) As seen in FIGS. **41** and **42**, as threaded nut **352** translates along threaded driver **357**, the first end of clevis **350** pivots about pins **352a** and mounting protrusion **312e**, relative to threaded nut **352**, such that the second end of clevis **350** moves axial drive cable **360** between an extended position (FIG. **42**) and a retracted position (FIG. **41**), as indicated by arrows “ZZ” and “YY.” The second end of clevis **350** also pivot relative to cable pivot **350** as axial drive cable **360** is moved between the extended and retracted positions to impart axial force to end effector **306**.

(71) The disclosed structure can include any suitable mechanical, electrical, and/or chemical components for operating the disclosed system or components thereof. For instance, such electrical components can include, for example, any suitable electrical and/or electromechanical, and/or electrochemical circuitry, which may include or be coupled to one or more printed circuit boards. As appreciated, the disclosed computing devices (and/or servers) can include, for example, a

“controller,” “processor,” “digital processing device” and like terms, and which are used to indicate a microprocessor or central processing unit (CPU). The CPU is the electronic circuitry within a computer that carries out the instructions of a computer program by performing the basic arithmetic, logical, control and input/output (I/O) operations specified by the instructions, and by way of non-limiting examples, include server computers. In some aspects, the controller includes an operating system configured to perform executable instructions. The operating system is, for example, software, including programs and data, which manages hardware of the disclosed apparatus and provides services for execution of applications for use with the disclosed apparatus. Those of skill in the art will recognize that suitable server operating systems include, by way of non-limiting examples, FreeBSD, OpenBSD, NetBSD®, Linux, Apple® Mac OS X Server®, Oracle® Solaris®, Windows Server®, and Novell® NetWare®. In some aspects, the operating system is provided by cloud computing.

(72) In some aspects, the term “controller” may be used to indicate a device that controls the transfer of data from a computer or computing device to a peripheral or separate device and vice versa, and/or a mechanical and/or electromechanical device (e.g., a lever, knob, etc.) that mechanically operates and/or actuates a peripheral or separate device.

(73) In aspects, the controller includes a storage and/or memory device. The storage and/or memory device is one or more physical apparatus used to store data or programs on a temporary or permanent basis. In some aspects, the controller includes volatile memory and requires power to maintain stored information. In various aspects, the controller includes non-volatile memory and retains stored information when it is not powered. In some aspects, the non-volatile memory includes flash memory. In certain aspects, the non-volatile memory includes dynamic random-access memory (DRAM). In some aspects, the non-volatile memory includes ferroelectric random-access memory (FRAM). In various aspects, the non-volatile memory includes phase-change random access memory (PRAM). In certain aspects, the controller is a storage device including, by way of non-limiting examples, CD-ROMs, DVDs, flash memory devices, magnetic disk drives, magnetic tapes drives, optical disk drives, and cloud-computing-based storage. In various aspects, the storage and/or memory device is a combination of devices such as those disclosed herein.

(74) In various aspects, the memory can be random access memory, read-only memory, magnetic disk memory, solid state memory, optical disc memory, and/or another type of memory. In various aspects, the memory can be separate from the controller and can communicate with the processor through communication buses of a circuit board and/or through communication cables such as serial ATA cables or other types of cables. The memory includes computer-readable instructions that are executable by the processor to operate the controller. In various aspects, the controller may include a wireless network interface to communicate with other computers or a server. In aspects, a storage device may be used for storing data. In various aspects, the processor may be, for example, without limitation, a digital signal processor, a microprocessor, an ASIC, a graphics processing unit (“GPU”), field-programmable gate array (“FPGA”), or a central processing unit (“CPU”).

(75) The memory stores suitable instructions and/or applications, to be executed by the processor, for receiving the sensed data (e.g., sensed data from camera), accessing storage device of the controller, generating a raw image based on the sensed data, comparing the raw image to a calibration data set, identifying an object based on the raw image compared to the calibration data set, transmitting object data to a post-processing unit, and displaying the object data to a graphic user interface. Although illustrated as part of the disclosed structure, it is also contemplated that a controller may be remote from the disclosed structure (e.g., on a remote server), and accessible by the disclosed structure via a wired or wireless connection. In aspects where the controller is remote, it is contemplated that the controller may be accessible by, and connected to, multiple structures and/or components of the disclosed system.

(76) The term “application” may include a computer program designed to perform functions, tasks, or activities for the benefit of a user. Application may refer to, for example, software running

locally or remotely, as a standalone program or in a web browser, or other software which would be understood by one skilled in the art to be an application. An application may run on the disclosed controllers or on a user device, including for example, on a mobile device, an IOT device, or a server system.

(77) In some aspects, the controller includes a display to send visual information to a user. In various aspects, the display is a cathode ray tube (CRT). In various aspects, the display is a liquid crystal display (LCD). In certain aspects, the display is a thin film transistor liquid crystal display (TFT-LCD). In aspects, the display is an organic light emitting diode (OLED) display. In certain aspects, on OLED display is a passive-matrix OLED (PMOLED) or active-matrix OLED (AMOLED) display. In aspects, the display is a plasma display. In certain aspects, the display is a video projector. In various aspects, the display is interactive (e.g., having a touch screen) that can detect user interactions/gestures/responses and the like. In some aspects, the display is a combination of devices such as those disclosed herein.

(78) The controller may include or be coupled to a server and/or a network. As used herein, the term “server” includes “computer server,” “central server,” “main server,” and like terms to indicate a computer or device on a network that manages the disclosed apparatus, components thereof, and/or resources thereof. As used herein, the term “network” can include any network technology including, for instance, a cellular data network, a wired network, a fiber-optic network, a satellite network, and/or an IEEE 802.11a/b/g/n/ac wireless network, among others.

(79) In various aspects, the controller can be coupled to a mesh network. As used herein, a “mesh network” is a network topology in which each node relays data for the network. All mesh nodes cooperate in the distribution of data in the network. It can be applied to both wired and wireless networks. Wireless mesh networks can be considered a type of “Wireless ad hoc” network. Thus, wireless mesh networks are closely related to Mobile ad hoc networks (MANETs). Although MANETs are not restricted to a specific mesh network topology, Wireless ad hoc networks or MANETs can take any form of network topology. Mesh networks can relay messages using either a flooding technique or a routing technique. With routing, the message is propagated along a path by hopping from node to node until it reaches its destination. To ensure that all its paths are available, the network must allow for continuous connections and must reconfigure itself around broken paths, using self-healing algorithms such as Shortest Path Bridging. Self-healing allows a routing-based network to operate when a node breaks down or when a connection becomes unreliable. As a result, the network is typically quite reliable, as there is often more than one path between a source and a destination in the network. This concept can also apply to wired networks and to software interaction. A mesh network whose nodes are all connected to each other is a fully connected network.

(80) In some aspects, the controller may include one or more modules. As used herein, the term “module” and like terms are used to indicate a self-contained hardware component of the central server, which in turn includes software modules. In software, a module is a part of a program. Programs are composed of one or more independently developed modules that are not combined until the program is linked. A single module can contain one or several routines, or sections of programs that perform a particular task.

(81) As used herein, the controller includes software modules for managing various aspects and functions of the disclosed system or components thereof.

(82) The disclosed structure may also utilize one or more controllers to receive various information and transform the received information to generate an output. The controller may include any type of computing device, computational circuit, or any type of processor or processing circuit capable of executing a series of instructions that are stored in memory. The controller may include multiple processors and/or multicore central processing units (CPUs) and may include any type of processor, such as a microprocessor, digital signal processor, microcontroller, programmable logic device (PLD), field programmable gate array (FPGA), or the like. The controller may also include

a memory to store data and/or instructions that, when executed by the one or more processors, cause the one or more processors to perform one or more methods and/or algorithms.

(83) The phrases “in an aspect,” “in aspects,” “in various aspects,” “in some aspects,” or “in other aspects” may each refer to one or more of the same or different aspects in accordance with the present disclosure. A phrase in the form “A or B” means “(A), (B), or (A and B).” A phrase in the form “at least one of A, B, or C” means “(A); (B); (C); (A and B); (A and C); (B and C); or (A, B, and C).”

(84) Various aspects disclosed herein may be combined in different combinations than the combinations specifically presented in the description and accompanying drawings. It should also be understood that, depending on the example, certain acts or events of any of the processes or methods described herein may be performed in a different sequence, may be added, merged, or left out altogether (e.g., all described acts or events may not be necessary to carry out the techniques).

(85) Certain aspects of the present disclosure may include some, all, or none of the above advantages and/or one or more other advantages readily apparent to those skilled in the art from the drawings, descriptions, and claims included herein. Moreover, while specific advantages have been enumerated above, the various aspects of the present disclosure may include all, some, or none of the enumerated advantages and/or other advantages not specifically enumerated above.

(86) The aspects disclosed herein are examples of the disclosure and may be embodied in various forms. For instance, although certain aspects herein are described as separate, each of the aspects herein may be combined with one or more of the other aspects herein. Specific structural and functional details disclosed herein are not to be interpreted as limiting, but as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the present disclosure in virtually any appropriately detailed structure. Like reference numerals may refer to similar or identical elements throughout the description of the figures.

(87) Any of the herein described methods, programs, algorithms, or codes may be converted to, or expressed in, a programming language or computer program. The terms “programming language” and “computer program,” as used herein, each include any language used to specify instructions to a computer, and include (but is not limited to) the following languages and their derivatives: Assembler, Basic, Batch files, BCPL, C, C+, C++, Delphi, Fortran, Java, JavaScript, machine code, operating system command languages, Pascal, Perl, PL1, scripting languages, Visual Basic, metalanguages which themselves specify programs, and all first, second, third, fourth, fifth, or further generation computer languages. Also included are database and other data schemas, and any other meta-languages. No distinction is made between languages which are interpreted, compiled, or use both compiled and interpreted approaches. No distinction is made between compiled and source versions of a program. Thus, reference to a program, where the programming language could exist in more than one state (such as source, compiled, object, or linked) is a reference to all such states. Reference to a program may encompass the actual instructions and/or the intent of those instructions.

(88) Securement of any of the components of the disclosed devices may be effectuated using known securement techniques such welding, crimping, gluing, fastening, etc.

(89) Persons skilled in the art will understand that the structures and methods specifically described herein and shown in the accompanying figures are non-limiting exemplary aspects, and that the description, disclosure, and figures should be construed merely as exemplary of aspects. It is to be understood, therefore, that this disclosure is not limited to the precise aspects described, and that various other changes and modifications may be effectuated by one skilled in the art without departing from the scope or spirit of the disclosure. Additionally, the elements and features shown or described in connection with certain aspects may be combined with the elements and features of certain other aspects without departing from the scope of this disclosure, and that such modifications and variations are also included within the scope of this disclosure. Accordingly, the subject matter of this disclosure is not limited by what has been particularly shown and described.

Claims

1. A robotic surgical system, comprising: a drive unit; a surgical instrument removably connected to the drive unit, the surgical instrument including: an elongated shaft assembly having a proximal end portion and a distal end portion; an end effector supported on the distal end portion of the elongated shaft assembly; and an instrument cassette assembly supported on the proximal end portion of the elongated shaft assembly, the instrument cassette assembly including: a cassette housing; and an actuator system supported in the cassette housing and operably coupled to the end effector for operating the end effector, the actuator system including: a cable actuator assembly including a plurality of cables that extends from the cassette housing to the end effector for manipulating the end effector; an actuator shaft assembly supported in the elongated shaft assembly and defining a longitudinal axis, the actuator shaft assembly including a first inner shaft, and a second inner shaft slidably advanceable through the first inner shaft; a rotation actuator assembly coupled to the first inner shaft of the actuator shaft assembly and positioned to rotate the first inner shaft of the actuator shaft assembly about the longitudinal axis and relative to the elongated shaft assembly for imparting rotational force to the end effector; and an axial actuator assembly coupled to the second inner shaft of the actuator shaft assembly and positioned to axially translate the second inner shaft of the actuator shaft assembly relative to the longitudinal axis and relative to the elongated shaft assembly for imparting axial force to the end effector.
2. The robotic surgical system of claim 1, wherein the cable actuator assembly includes a crank, a first slider, and a second slider, the first and second sliders coupled to the crank.
3. The robotic surgical system of claim 2, wherein the crank is rotatable to linearly translate the first and second sliders relative to one another.
4. The robotic surgical system of claim 3, wherein the first slider supports a first cable of the plurality of cables and the second slider supports a second cable of the plurality of cables.
5. The robotic surgical system of claim 3, wherein the crank is coupled to a driver that is engaged with the drive unit, the driver configured to impart rotational force on the crank.
6. The robotic surgical system of claim 1, wherein the rotation actuator assembly includes a drive wheel and a belt drive shaft supporting a belt, the belt coupled to the actuator shaft assembly and the drive wheel coupled to the belt drive shaft.
7. The robotic surgical system of claim 6, wherein the drive wheel and the belt drive are disposed transverse to one another, the drive wheel configured to rotate the belt drive shaft.
8. The robotic surgical system of claim 6, wherein rotation of the belt drive shaft rotates the belt to rotate the first inner shaft of the actuator shaft assembly.
9. The robotic surgical system of claim 1, wherein the axial actuator assembly includes a drive disc, a drive arm coupled to the drive disc, and a drive plate coupled to the drive arm and to the shaft assembly.
10. The robotic surgical system of claim 9, wherein the drive arm includes a first pin coupled to the drive disc and a second pin coupled to the drive plate, the drive plate defining a pin slot that receives the second pin, the second pin slidable along the pin slot to axially translate the drive plate and the second inner shaft of the actuator shaft assembly as the drive disc rotates.
11. A surgical system, comprising: a cassette housing; an elongated shaft assembly having a proximal end portion supported in the cassette housing, and a distal end portion supporting an end effector; and an actuator system supported in the cassette housing and the elongated shaft, the actuator system including: a cable actuator assembly including a plurality of cables; an actuator shaft assembly supported in the elongated shaft assembly and defining a longitudinal axis, the actuator shaft assembly including a first inner shaft, and a second inner shaft slidably advanceable through the first inner shaft; a rotation actuator assembly coupled to the first inner shaft of the actuator shaft assembly and positioned to rotate the first inner shaft of the actuator shaft assembly

about the longitudinal axis and relative to the elongated shaft assembly; and an axial actuator assembly coupled to the second inner shaft of the actuator shaft assembly and positioned to axially translate the second inner shaft of the actuator shaft assembly relative to the longitudinal axis and relative to the elongated shaft assembly.

12. The surgical system of claim 11, wherein the cable actuator assembly includes a crank, a first slider, and a second slider, the first and second sliders coupled to the crank.

13. The surgical system of claim 12, wherein the crank is rotatable to linearly translate the first and second sliders relative to one another.

14. The surgical system of claim 13, wherein the first slider supports a first cable of the plurality of cables and the second slider supports a second cable of the plurality of cables.

15. The surgical system of claim 13, wherein the crank is coupled to a driver, the driver configured to impart rotational force on the crank.

16. The surgical system of claim 11, wherein the rotation actuator assembly includes a drive wheel and a belt drive shaft supporting a belt, the belt coupled to the actuator shaft assembly and the drive wheel coupled to the belt drive shaft.

17. The surgical system of claim 16, wherein the drive wheel and the belt drive are disposed transverse to one another, the drive wheel configured to rotate the belt drive shaft.

18. The surgical system of claim 16, wherein rotation of the belt drive shaft rotates the belt to rotate the first inner shaft of the actuator shaft assembly.

19. The surgical system of claim 11, wherein the axial actuator assembly includes a drive disc, a drive arm coupled to the drive disc, and a drive plate coupled to the drive arm and to the actuator shaft assembly.

20. A surgical instrument for a robotic surgical system, the surgical instrument comprising: an elongated shaft assembly having a proximal end portion and a distal end portion; an end effector supported at the distal end portion of the elongated shaft assembly; and an instrument cassette assembly supported on the proximal end portion of the elongated shaft assembly, the instrument cassette assembly including: a cassette housing; and an actuator system supported in the cassette housing and operably coupled to the end effector for operating the end effector, the actuator system including: a cable actuator assembly including a plurality of cables that extends from the cassette housing to the end effector for manipulating the end effector; an actuator shaft assembly supported in the elongated shaft assembly and defining a longitudinal axis, the actuator shaft assembly including a first inner shaft, and a second inner shaft slidably advanceable through the first inner shaft; a rotation actuator assembly coupled to the first inner shaft of the actuator shaft assembly and positioned to rotate the first inner shaft of the actuator shaft assembly about the longitudinal axis and relative to the elongated shaft assembly for imparting rotational force to the end effector; and an axial actuator assembly coupled to the second inner shaft of the actuator shaft assembly and positioned to axially translate the second inner shaft of the actuator shaft assembly relative to the longitudinal axis and relative to the elongated shaft assembly for imparting axial force to the end effector.
